RIDE SHARE PRICE PREDICTOR

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DATASET & GOAL

lyft vs Uber

Dataset:

- Ride share data from Boston, Massachusetts from both Uber and Lyft
- Includes price, distance, destination, time, pricing factors, weather conditions, etc.
- Each row represents a single ride

Goal: Create a regression model to accurately predict the price of a ride

AGENDA

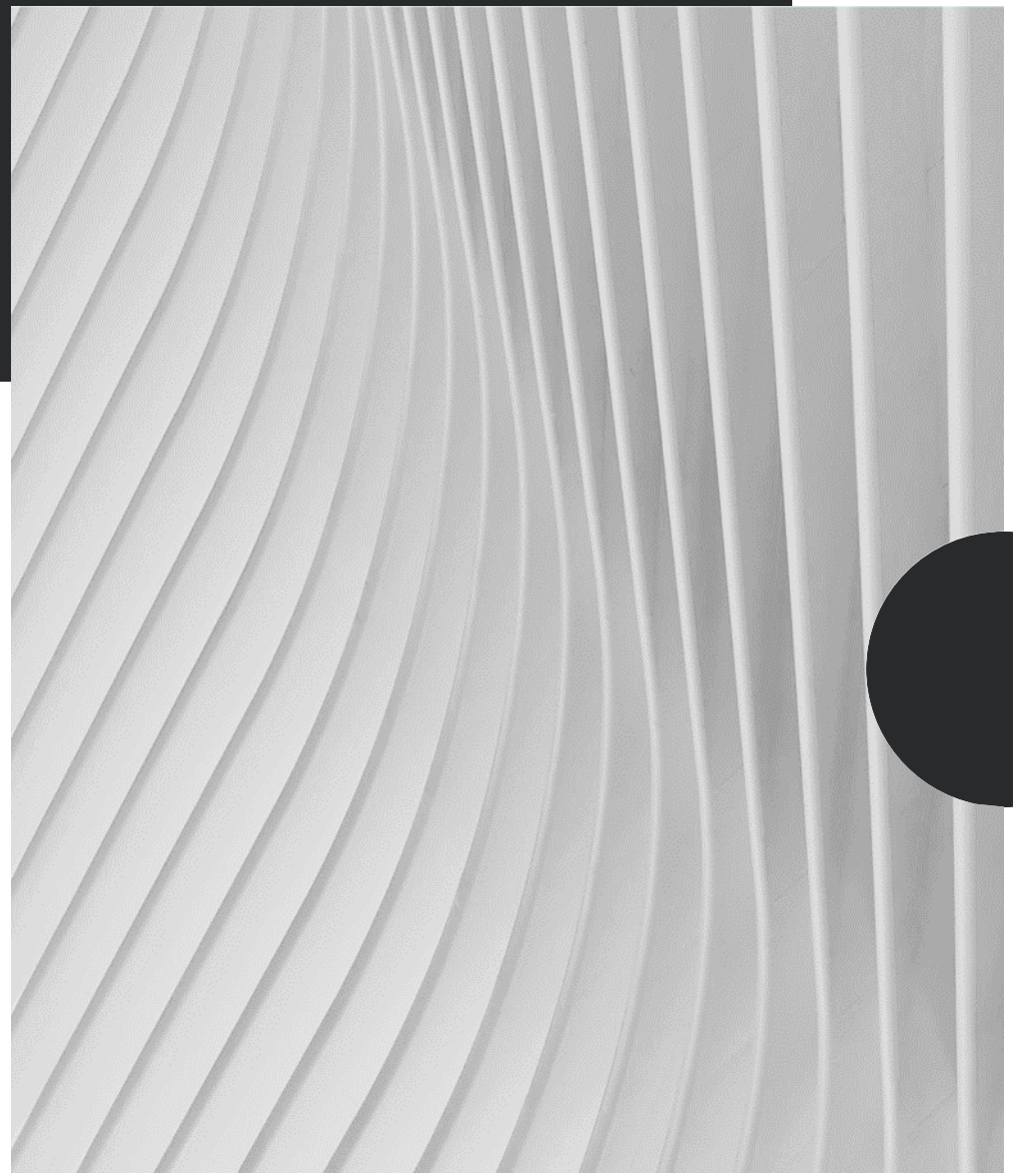
Cleaning & Preparation

Exploration

Creation

- Simple Linear
- Multiple
- Multiple with Higher Degree & Ridge Coefficient

Evaluation

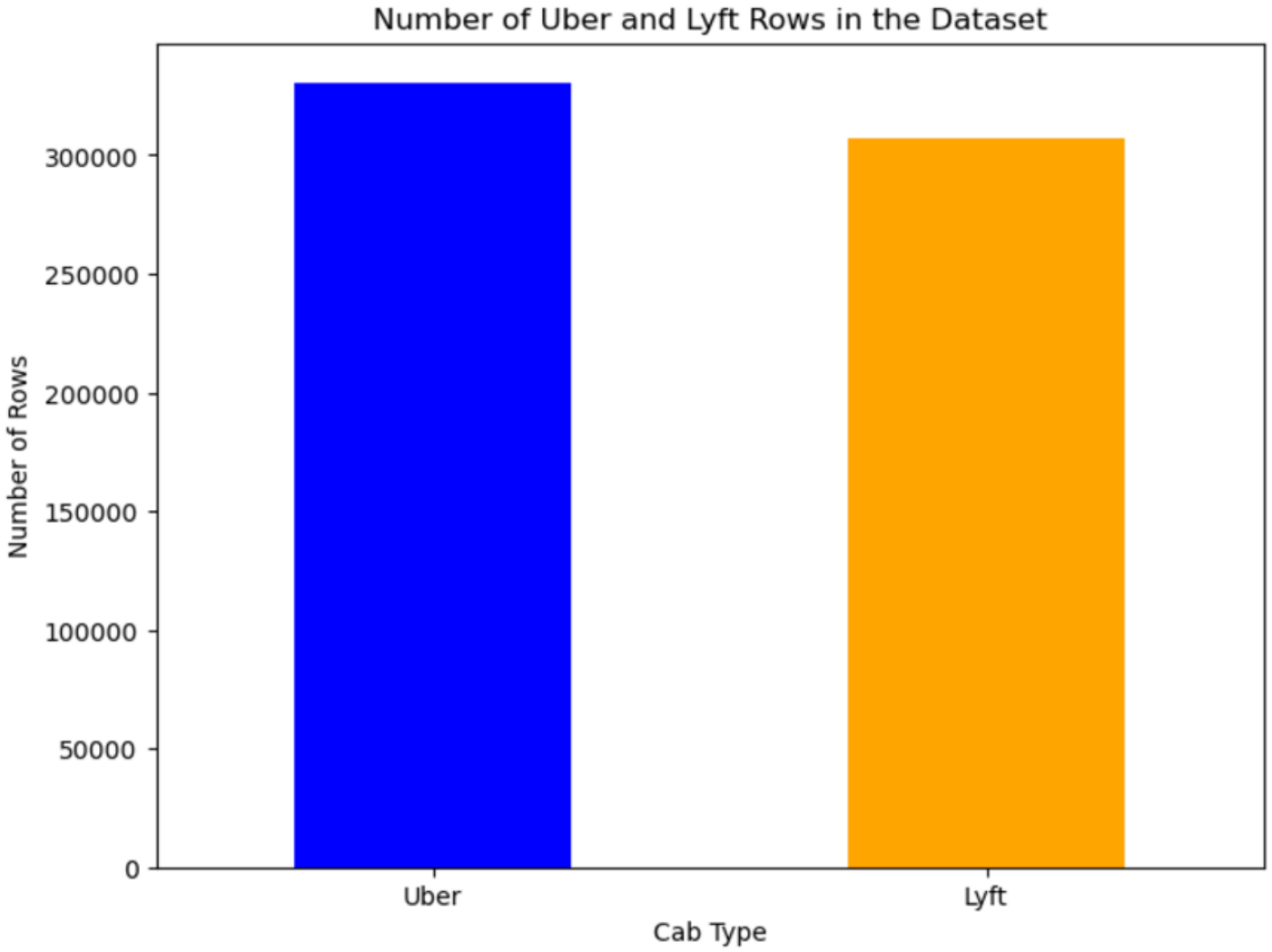


CLEANING & PREP

- 693,071 to 637,976 from dropping 55,095 null price records
- Encoding for Categorical Data, e.g. Uber, Lyft

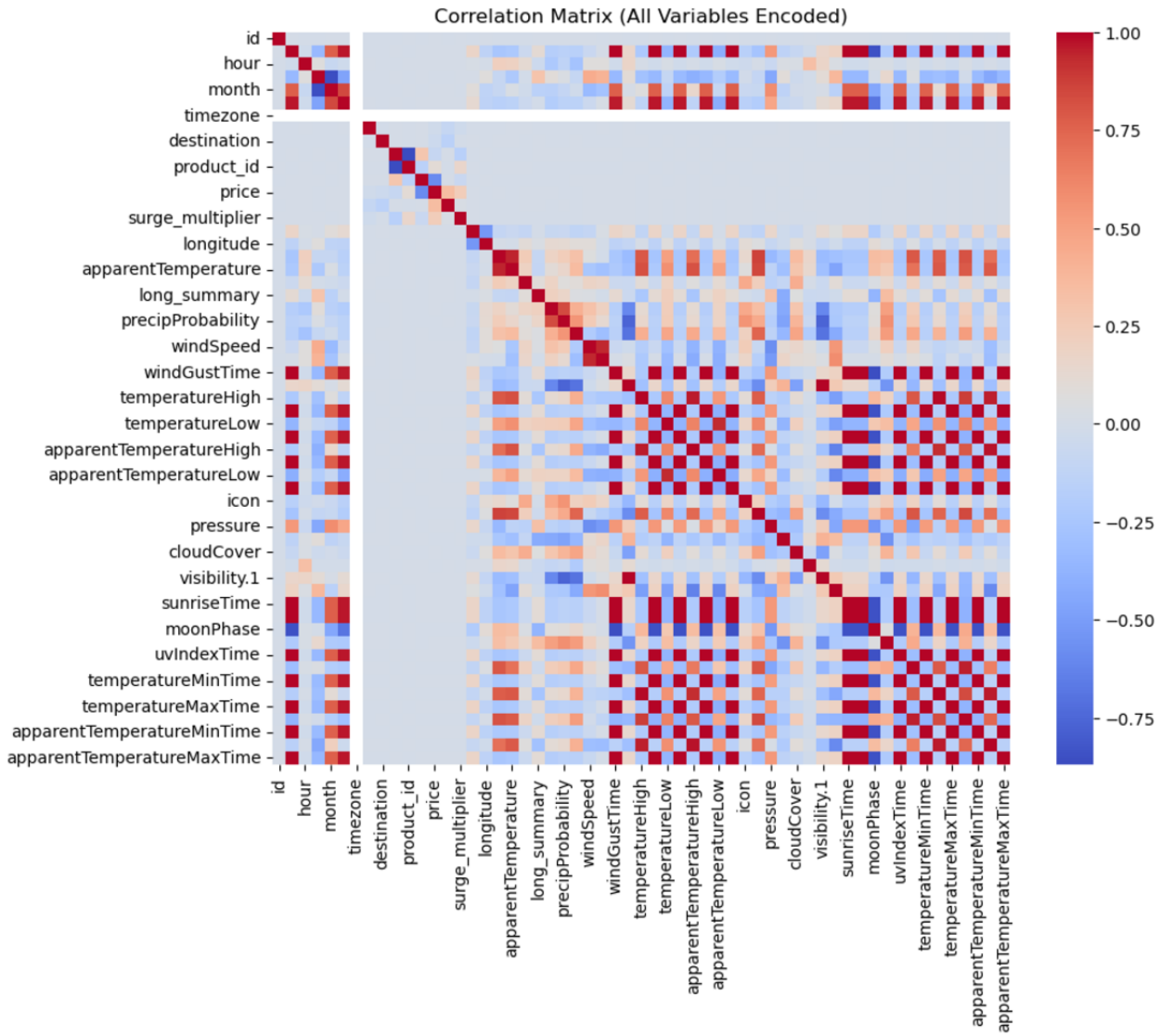
DATA EXPLORATION

RECORD DISTRIBUTION



PRICE V. DISTANCE

CORRELATION MATRIX

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Variables correlated to price:
 price                          1.000000
distance                        0.345061
surge_multiplier                0.240458
product_id                      0.102679
icon                            0.002100
latitude                        0.002088
visibility.1                    0.001497
visibility                      0.001497
short_summary                   0.001245
windGust                        0.001218
id                              0.001137
precipIntensityMax              0.001114
windSpeed                       0.000990
month                           0.000913

dewPoint                       -0.000615
day                            -0.000722
humidity                       -0.001238
windBearing                    -0.001382
longitude                      -0.001417
moonPhase                      -0.001602
source                         -0.025366
destination                    -0.046320
cab_type                       -0.083385
name                           -0.586644
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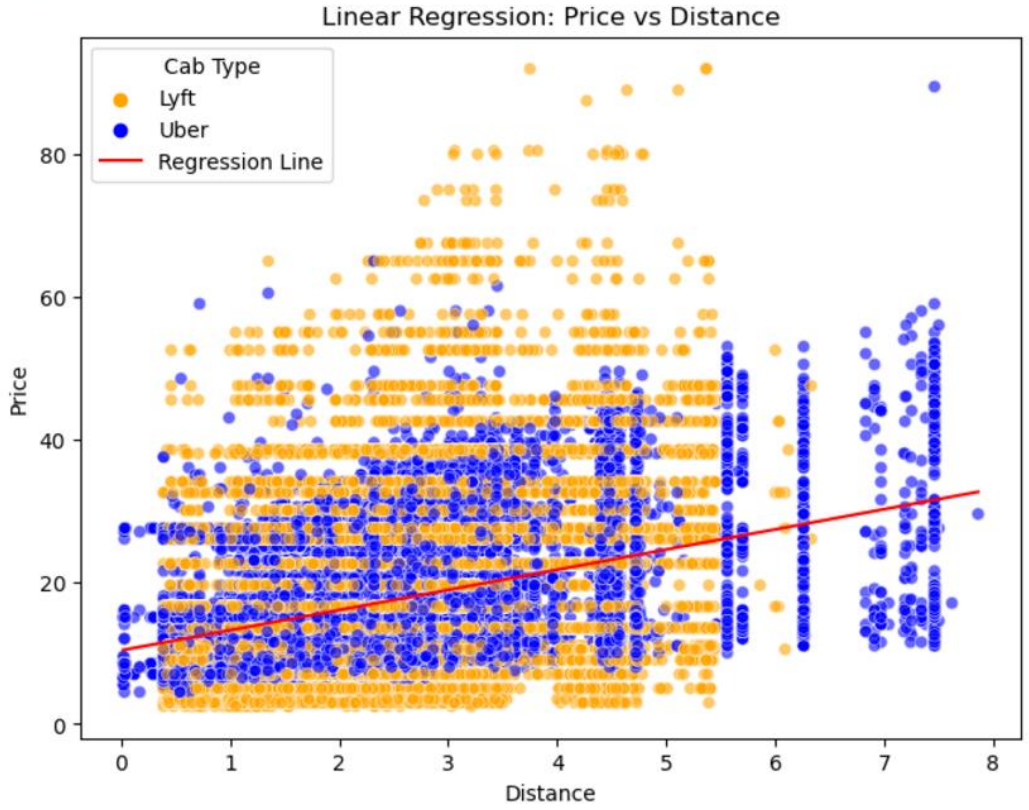
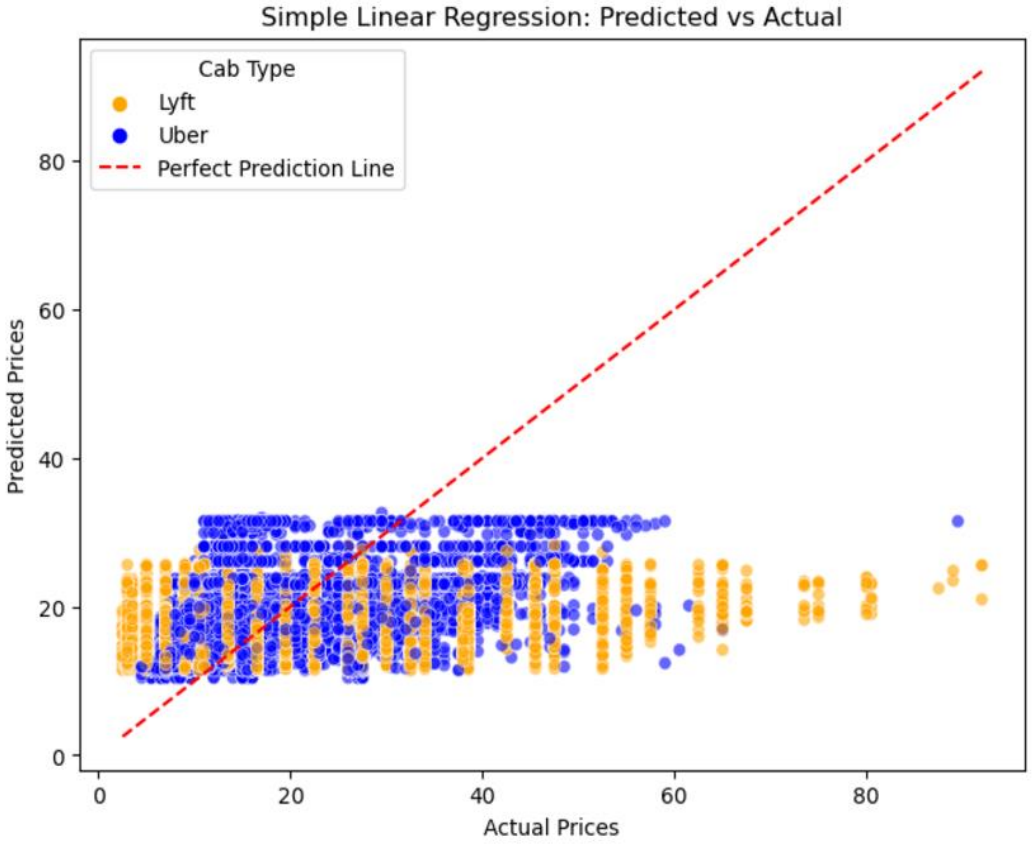
MODEL CREATION & EVALUATION

SIMPLE - DISTANCE

R-squared (R2): 0.11940541734213972

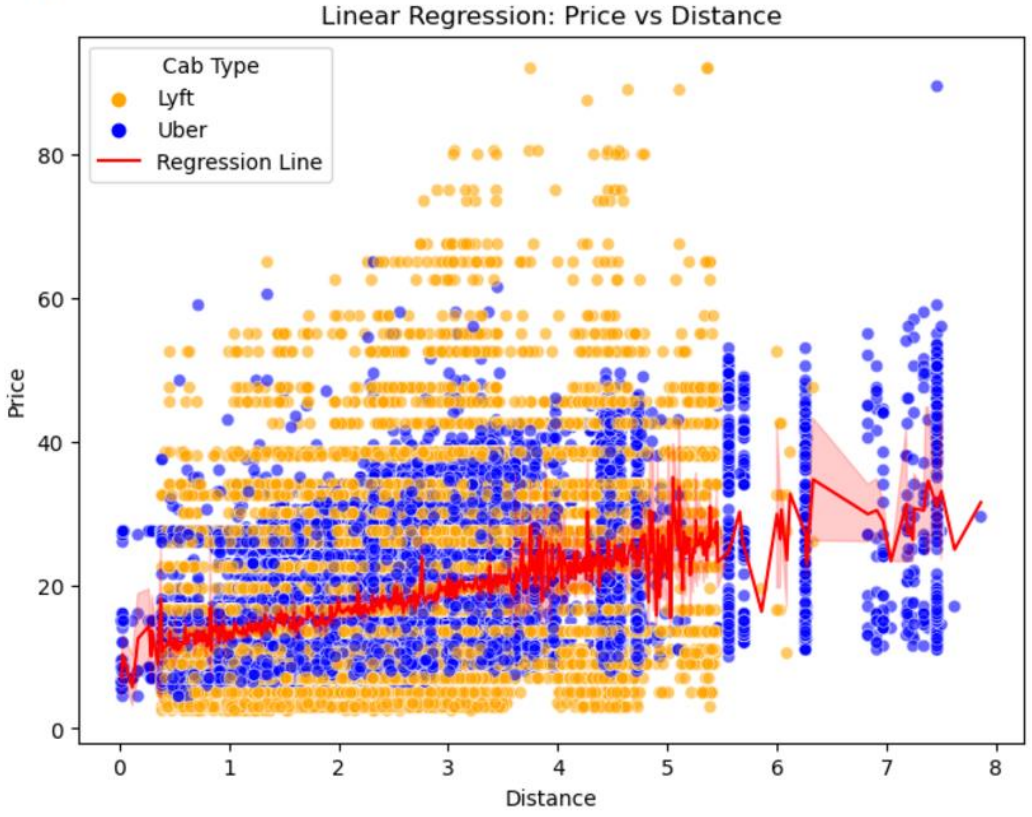


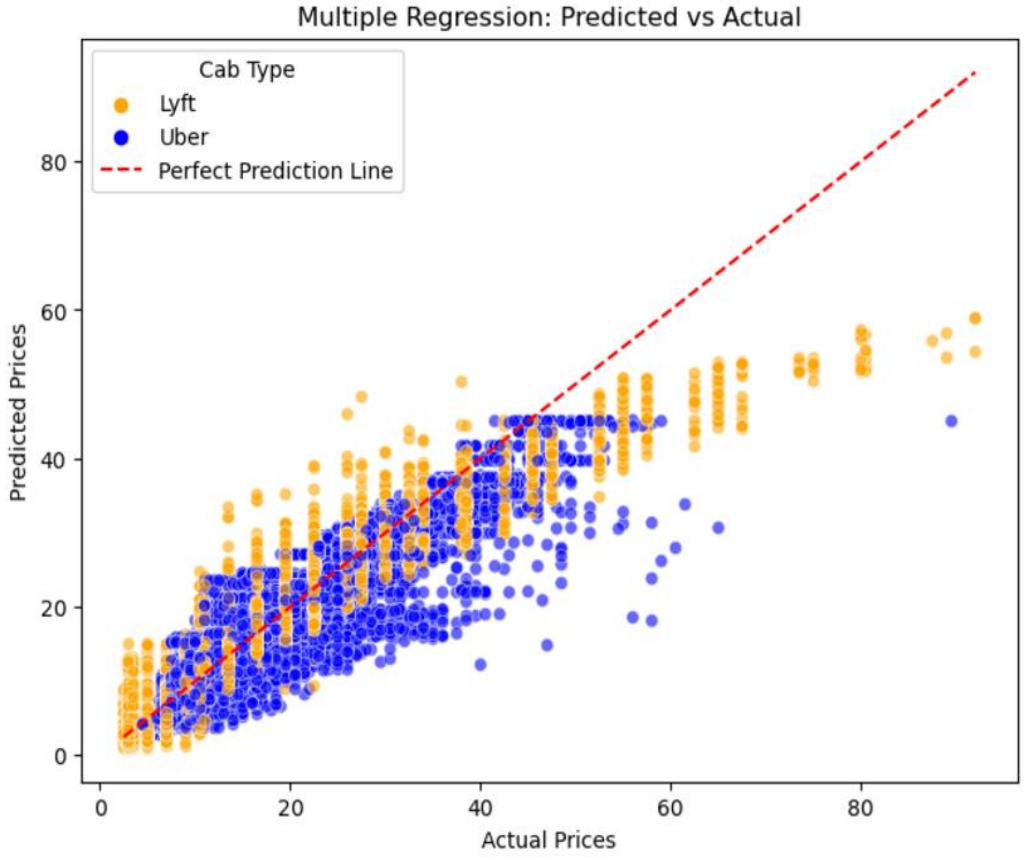


MULTIPLE REGRESSION

DISTANCE, SURGE MULTIPLIER, PRODUCT ID

R-squared (R^2): 0.9273953169002663





POLYNOMIAL REGRESSION

DISTANCE, SURGE MULTIPLIER, PRODUCT ID

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Optimal Polynomial Degree: 3
Optimal Ridge Alpha: 0.1
Polynomial + Ridge Regression - R-squared (R²): 0.9611540885615378
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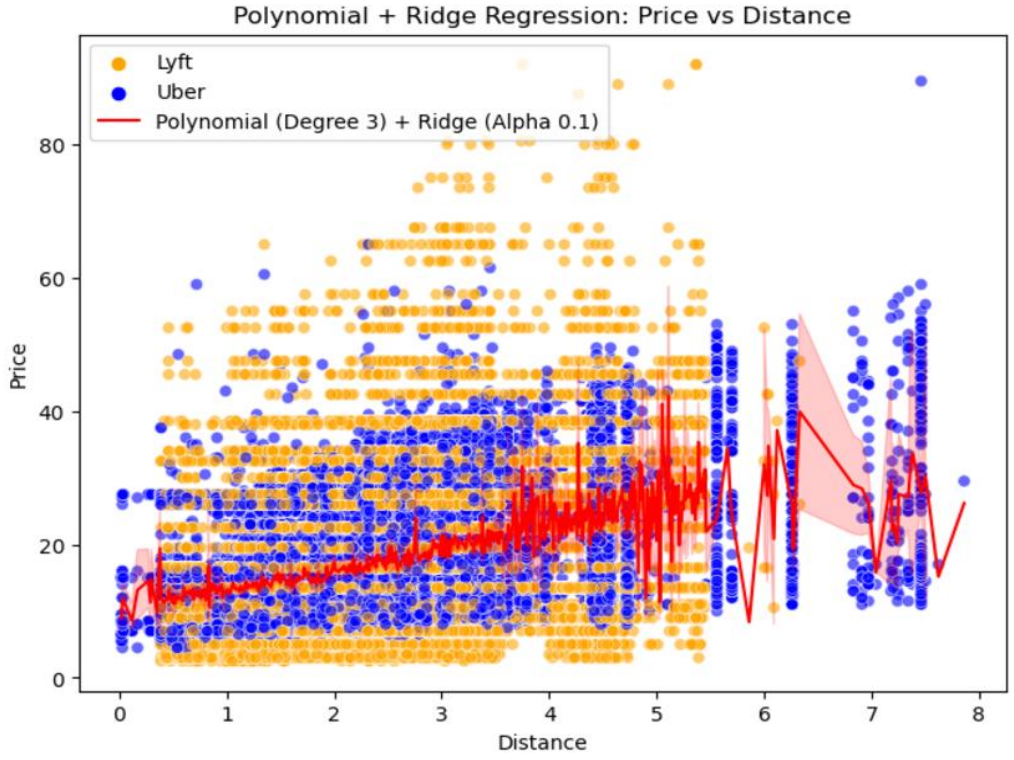

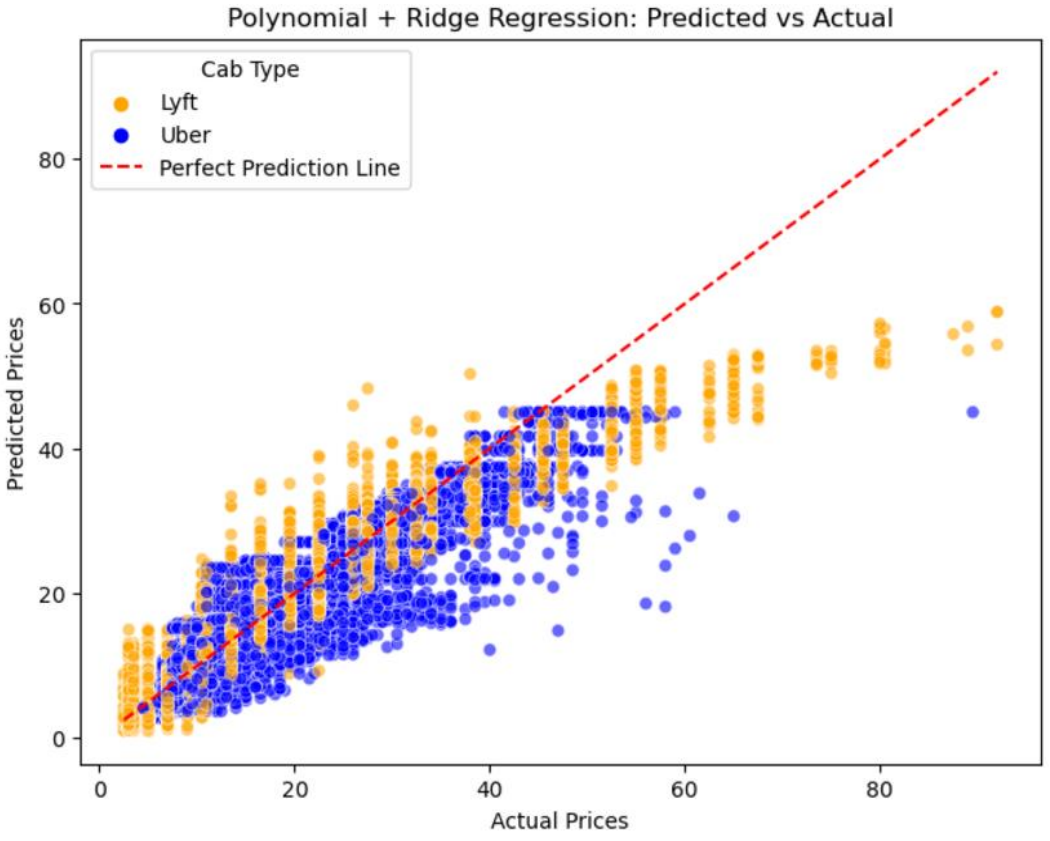

THANK YOU

Questions?